

Administrator Manual — Tracker Porti

🇮🇹 Manuale in italiano: [manuale_admin.md](#) · [index.html](#)

Guide to the functions reserved for Tracker Porti **administrators**: user and group management, coverage-map controls, risk model, system logs, and editing configuration files.

This manual **complements** the [User manual](#), which covers day-to-day use of the app. Here you'll find **only** what is reserved for administrators. For common features (monitoring, followed ships, ship detail, notifications, export...) refer to the user manual.

Administrator role

Amministrazione utenti

← App · Esci

NOME	EMAIL / USERNAME	RUOLO	STATO	ONLINE	GRUPPO	AREE	AZIONI
Amministratore (tu)	admin@local admin	admin	active	●	—	1	Disabilita Rendi utente Reset password
Manuale Utente	manuale.utente@tracker-porti.local manuale_utente	user	active	●	Monitoraggio Toscana	1	Disabilita Rendi admin Reset password Impersona Elimina
Squadra Porto	squadra.porto@tracker-porti.local squadra_porto	user	active	●	Monitoraggio Toscana	1	Disabilita Rendi admin Reset password Impersona Elimina

Gruppi di utenti

I membri di un gruppo condividono (unione) aree di monitoraggio, navi seguite, navi flaggate, navi silenziate e le preferenze di notifica/mappa. Le impostazioni gestite dall'amministratore (sorgenti dati, pesi di rischio) restano globali. Un gruppo richiede almeno 2 utenti.

+ Nuovo gruppo

NOME	DESCRIZIONE	MEMBRI	AZIONI
		Manuale Utente	Squadra Porto

Administration page: user table with role, status, group, areas, and available actions.

Administrators see the **Admin** link at the top right, which opens the **administration page** (/admin). An administrator can:

- **approve** pending new registrations;

- **enable or disable** an account;
- **change a user's role** (regular user ↔ administrator ↔ tester);
- **reset a user's password** — generates a **one-time link** (valid 24 hours) to hand to them;
- **delete** a user;
- **create and manage user groups** (see below);
- **impersonate** a user — view their areas, monitoring, and followed ships in **read-only** mode, with a prominent banner and one-click exit;
- consult the **logs** (activity log and API log), shared and visible only to administrators.

Actions on a user row

Each table row shows **at most one highlighted button** — the one that row is waiting for: **Approve** for a pending registration, **Re-enable** for a disabled account. Everything else lives in the ... menu at the end of the row, always in the same order:

1. **status** — Approve / Approve as tester / Disable / Re-enable;
2. **role** — Make administrator, Revoke administrator, or **Promote to user** (for a tester account);
3. **utilities** — Reset password, Impersonate;
4. **destructive action** — Delete user (in red, with confirmation).

From tester to regular user

The **tester** role is only assigned at approval time, with **Approve as tester**: it is an account with reduced limits (number and size of areas, number of followed ships) meant for trials.

When you want to lift those limits, open the ... menu on the user's row and choose **Promote to user**: the account becomes a regular user, with no tester limits, keeping the areas, followed ships and settings already configured. The move can only be reversed through a new approval, so the tester role cannot be re-assigned to an account that has been promoted.

Settings managed by administrators (data sources, sanctions/PSC screening, risk-score weights) are **global** for all users. In the Settings interface, the tabs and toggles reserved for administrators are **hidden** from regular users (and protected server-side regardless).

User groups

Amministrazione utenti

← App · Esci

Amministratore (tu)	admin@local admin	admin	active	—	1	Rendi utente Reset password
Manuale Utente	manuale.utente@tracker-porti.local manuale_utente	user	active	●	Monitoraggio Toscana 1	Disabilita Rendi admin Reset password Impersona Elimina
Squadra Porto	squadra.porto@tracker-porti.local squadra_porto	user	active	●	Monitoraggio Toscana 1	Disabilita Rendi admin Reset password Impersona Elimina

Gruppi di utenti

I membri di un gruppo condividono (unione) aree di monitoraggio, navi seguite, navi flaggate, navi silenziate e le preferenze di notifica/mappa. Le impostazioni gestite dall'amministratore (sorgenti dati, pesi di rischio) restano globali. Un gruppo richiede almeno 2 utenti.

+ Nuovo gruppo

NOME	DESCRIZIONE	MEMBRI	AZIONI
Monitoraggio Toscana	Squadra che segue il traffico costiero toscano	Manuale Utente Squadra Porto + aggiungi membro...	Rinomina Sciogli

Administration page: User groups section with the creation form and the list of existing groups.

An administrator can put several users into a **group**. When a user is part of a group, they **share with the other members** (as a **union** of what each one had):

- monitoring **areas**;
- **followed ships**, **flagged ships** ★, and **muted ships** 🚫;
- **notification preferences** (app and Telegram) and **map display**, plus the **default area**.

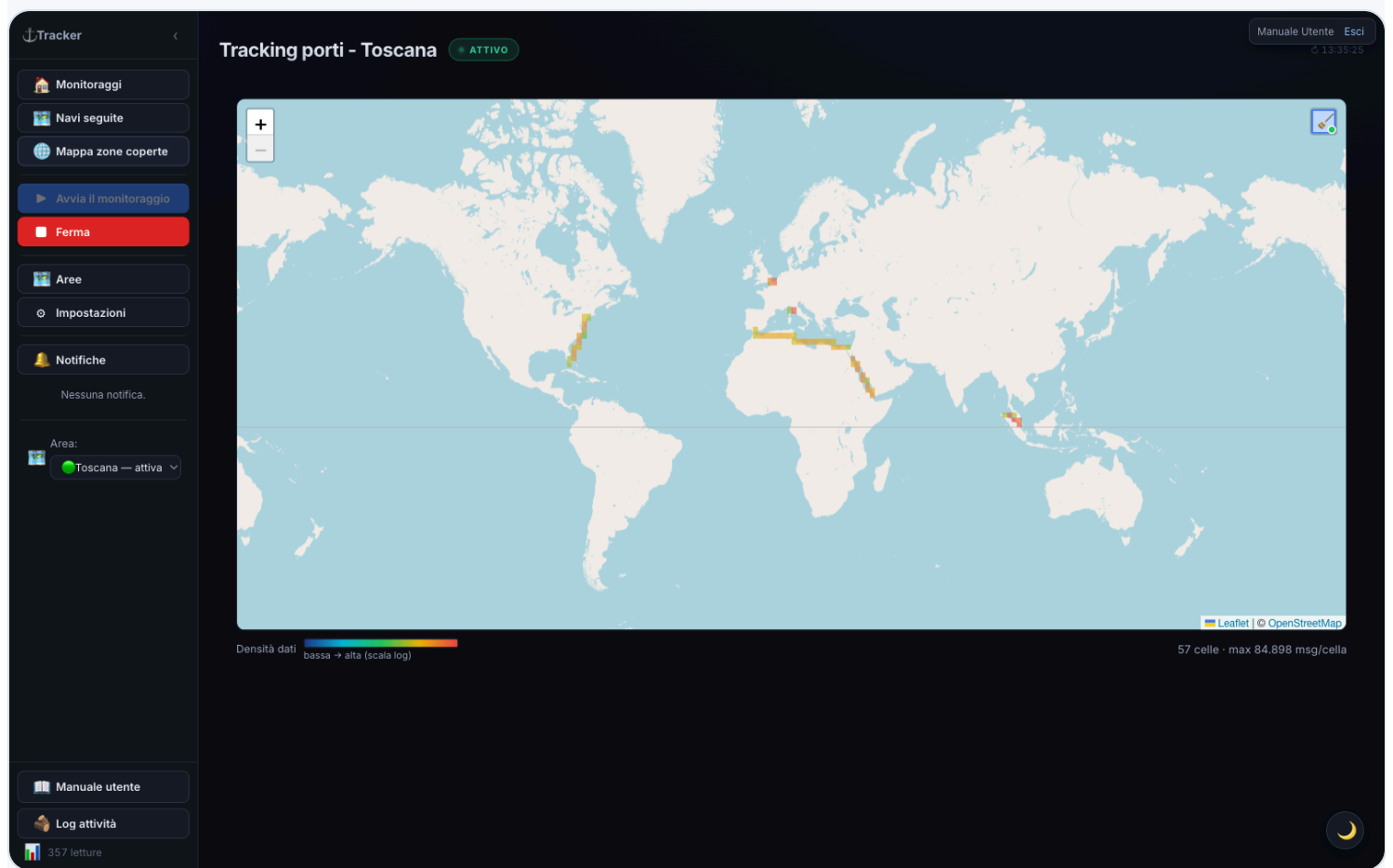
In practice: if one member adds an area, follows a ship, or enables a notification, **the other members get it** on their next visit — and vice versa (write-through sync). What stays **personal** to each user: the **Telegram link** (their own chat) and the interface **language**.

Management (from the Admin page): a group has a **name**, a **description**, and **at least 2 members**. On creation you choose the **template user** whose settings the group starts from. You can **add/remove** members or **dissolve** the group.

A removal that would leave **only 1 member** is blocked: in that case, **dissolve** the group instead. If an administrator removes a user from a group, that user **keeps** everything that was shared up to that point (areas, ships, settings): they simply stop syncing with the others.

Group activity log: every write-through share is also recorded (who, when, what) — members review it themselves in the Group activity section (visible to anyone in a group, no admin action needed). Rows older than `GROUP_ACTIVITY_LOG_RETENTION_DAYS` days (default 90, in `app.config.properties`) are deleted automatically.

Coverage map — administrator controls



World coverage map: grid colored by AIS message density.

The Coverage map is visible read-only to all users (and even without login at `/heatmap`). Only **administrators** can additionally:

- **start** and **stop** data collection (**Start/Stop collection** buttons);
- see **real-time connection statistics** (bandwidth used, messages per second, populated cells...);
- **Refresh map** and **Clear data** (empties the collected grid).

How collection works: once started, it runs **in the background** on the server — even if no one has the page open — until an administrator stops it. It resumes on its own after a server restart. As a safety measure, it **automatically shuts off** if no user has been active for **10 minutes**.

Warning: while collection is active, the app **downloads data from the whole world** continuously (roughly **200–400 MB per hour**). The feature requires an **AISStream key from a separate account** (different from the one used for regular monitoring), set in `local.properties` as `HEATMAP_AIS_API_KEY`. Without that key, the feature stays off.

Risk model (signal weights)

Besides the **cargo-type weights** (editable by everyone in Settings), administrators can adjust **how many points each risk signal is worth** (AIS blackout, spoofing/position jump, dwell, draught increase, sanctions, PSC, GFW events, rendezvous, etc.) from ⚙ **Settings** → "⚙ **Risk model**" section.

- A **grid** with one field per signal: change the values and press  **Save weights. Immediate** effect, no restart needed. **Reset to default** restores factory values.
- Setting a weight to **0** disables that signal.
- You can save full configurations as **risk profiles** (*Risk profile* menu → *Save as...*) and recall them with *Apply* — useful for quickly switching between different setups (e.g. a more aggressive profile, or one tuned for areas with poor AIS coverage).

Detection thresholds and signal **multipliers** are not in the UI: they stay in the `app.config.properties` file (`RISK_*` keys).

The Settings **General** tab, "⚙ Risk model" section, also has two toggles **reserved for administrators** (invisible to other users):

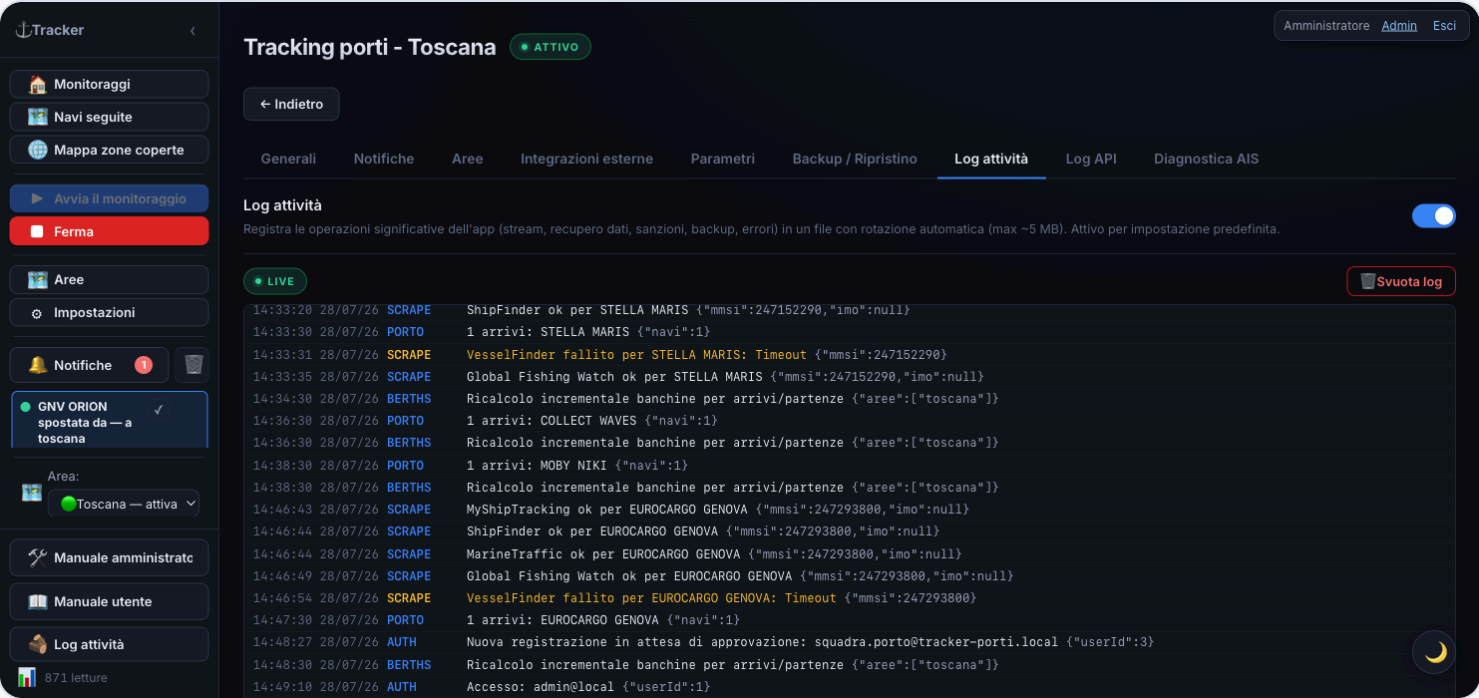
- **Exclude tankers** — doesn't assign the "ship type" score to tanker hulls (AIS code 80–89). Useful when monitoring weapons transport, which a tanker can't carry out: it only zeroes that one factor, not the whole score (a tanker can still land in the red band from other signals — sanctions, AIS dark, etc.). It's a **global** value shared by every user, and different from the notifications' ship-type filter (Settings → **Notifications**, personal to each user): that one only decides what reaches you as a notification, without touching the score.

- **Check position jump / Check AIS blackout** — include those signals in the score. Disable them in areas with poor AIS coverage, where sparse reports produce false positives (apparent jumps/gaps that aren't real).

System logs

Logs are **shared** and visible **only to administrators**, as tabs in Settings.

Activity log



Settings — Activity log: chronological record of the app's operations.

Records the app's significant operations (streams, data fetching, sanctions, backups, errors) in a file with **automatic rotation** (max ~5 MB). On by default (**Activity log** toggle in the General tab, admin). The log can also be viewed from the 🗑️ **Activity log** overlay in the sidebar and can be **cleared**.

The log also records the **area-change notifications discarded** by the filters (Area change not notified: ...) with the reason: overlapping areas , only crossed , call too old . That is where to look when a user reports a missing alert they expected — see the AREA_CHANGE_* parameters in `app.config.properties` .

API log

Tracker

Monitoraggi

Navi seguite

Mappa zone coperte

Avvia il monitoraggio

Ferma

Aree

Impostazioni

Notifiche

GNV ORION spostata da — a toscana

Area: Toscana — attiva

Manuale amministratore

Manuale utente

Log attività

871 letture

Tracking porti - Toscana

ATTIVO

← Indietro

Generali

Notifiche

Aree

Integrazioni esterne

Parametri

Backup / Ripristino

Log attività

Log API

Diagnostica AIS

LIVE

Auto-scroll

Cancella log

#	ORA	METODO	PATH	STATUS	DURATA
29641	14:50:49 28/07/26	GET	/api/stream/status?lang=it	200 OK	0ms
29642	14:50:49 28/07/26	GET	/api/areas?lang=it	200 OK	2ms
29643	14:50:49 28/07/26	GET	/api/ships/past/count?area=toscana&lang=it	304 Not Modified	1ms
29644	14:50:54 28/07/26	DB	/ais/toscana/PositionReport	200 OK	1ms
29645	14:50:57 28/07/26	DB	/ais/toscana/ShipStaticData	200 OK	1ms
29646	14:51:05 28/07/26	GET	/api/app-log?limit=1000&lang=it	200 OK	44ms
29647	14:51:07 28/07/26	DB	/ais/toscana/PositionReport	200 OK	3ms
29648	14:51:10 28/07/26	DB	/ais/toscana/PositionReport	200 OK	0ms
29649	14:51:24 28/07/26	DB	/ais/toscana/StandardClassBPositionReport	200 OK	1ms
29651	14:51:31 28/07/26	AIS	/ais/monitoring/heartbeat	200 OK	0ms

Settings — API log: list of /api calls with method, path, and outcome.

Lists the application’s API calls in real time (method, path, status, time), useful for diagnostics. Sensitive request bodies (login, etc.) are **suppressed**: passwords are never logged.

AIS diagnostics

The screenshot displays the 'Tracking porti - Toscana' interface with the 'Diagnostica AIS' tab selected. The interface is divided into three main sections: 'Monitoraggio aree (toscana)', 'Navi seguite', and 'Mappa zone coperte'. Each section shows connection status, session uptime, frames received, message rate, reconnections, and the last error.

Monitoraggio aree (toscana)	Navi seguite	Mappa zone coperte
Chiave: _cf80 (local.properties) - local.properties	Chiave: _7cea (local.properties) - local.properties	Chiave: _960e (local.properties) - local.properties
CONNESSIONE ● Connesso 13:38:31 28/07/2026	CONNESSIONE ● Disconnesso	CONNESSIONE ● Disconnesso
UPTIME SESSIONE 17m 39s	UPTIME SESSIONE 11m 17s	UPTIME SESSIONE 0s
FRAME WS RICEVUTI 120	FRAME WS RICEVUTI 0	
MESSAGGI NAVE (SESSIONE) 120	MESSAGGI NAVE (SESSIONE) 0	
VELOCITÀ MESSAGGI 7 msg/min	VELOCITÀ MESSAGGI 0 msg/min	
TOTALE RICONNESSIONI 0	TOTALE RICONNESSIONI 0	
ULTIMO ERRORE AIS Nessuno	IN RI-ACQUISIZIONE 0	

Settings — AIS Diagnostics tab: connection status, uptime, frames received, and reconnections.

The **Settings → AIS Diagnostics** tab (visible only to administrators) shows the active area's data stream connection status, refreshed every 5s:

- **Connection** — Connected / Disconnected
- **Session uptime** — how long the stream has been active
- **WS frames received / Ship messages / Message rate**
- **Reconnections** — how many times the connection was restored
- **Last error** — the most recent error, if any

The **AIS outage banner** shown to all users on monitoring pages when an area goes quiet for a long time is described from the user's perspective in the [user manual](#); the external-confirmation mechanism behind it is in the [Credits](#) section below.

Editing configuration files

Some advanced settings aren't in the interface but in text files in the project folder. Open them with a text editor, change the values, and **restart the application** to apply them. Lines starting with `#` (or `//`) are comments and are ignored.

`local.properties` — keys and secrets

Contains the API key and initial preferences. Format `KEY=value`, one per line. **Must not be shared** (it contains the API key) and is in `.gitignore`. If it doesn't exist, copy it from `local.properties.example`.

Key	Meaning
<code>AIS_API_KEY</code>	AISStream.io key (required) — used by the monitoring areas' streams
<code>FOLLOW_AIS_API_KEY</code>	AISStream.io key for the followed ships stream. Best from a separate account (see note). Empty = reuses <code>AIS_API_KEY</code>
<code>HEATMAP_AIS_API_KEY</code>	Key of a separate AISStream account, used only for the Coverage map. Empty = feature disabled. Must be written "bare", with no comments on the same line
<code>BBOX_PRESET</code>	Area shown at startup (an area's key, e.g. <code>bari</code>)
<code>IMPORT_VF_DATA</code> / <code>IMPORT_MT_DATA</code>	<code>true</code> / <code>false</code> — enable VesselFinder / MarineTraffic import
<code>IMPORT_SF_DATA</code>	<code>true</code> / <code>false</code> — ShipFinder import + position to re-locate lost followed ships
<code>IMPORT_MST_DATA</code>	<code>true</code> / <code>false</code> — MyShipTracking import (second, independent backup position source)
<code>IMPORT_SANCTIONS</code>	<code>true</code> / <code>false</code> — screening against the OFAC SDN list
<code>IMPORT_SANCTIONS_EXTRA</code>	<code>true</code> / <code>false</code> — EU / UK OFSI / UN lists (only with <code>IMPORT_SANCTIONS</code>); default <code>true</code>
<code>IMPORT_PSC</code>	<code>true</code> / <code>false</code> — Port State Control screening (Paris/Tokyo MoU flags + banned ships)
<code>IMPORT_EQUASIS</code>	<code>true</code> / <code>false</code> — on-demand Equasis lookup in the ship detail
<code>EQUASIS_USER</code> / <code>EQUASIS_PASSWORD</code>	Equasis account credentials (free registration at equasis.org)
<code>IMPORT_GFW</code>	<code>true</code> / <code>false</code> — Global Fishing Watch enrichment; default <code>true</code>

Key	Meaning
GLOBAL_FISHING_WATCH_TOKEN	Global Fishing Watch API (Bearer) token. Data free for non-commercial use only
ADMIN_USERNAME / ADMIN_EMAIL / ADMIN_PASSWORD	Default administrator (created at startup if missing). ADMIN_PASSWORD is required: if it isn't configured and no admin exists yet, startup stops with an error instead of creating an account with a weak password. Change the password on a server reachable by others
COOKIE_SECURE	true / false — sends the session cookie only over HTTPS; set to true behind TLS
SESSION_TTL_DAYS	Login session length in days. Default 30

💡 **Recommended: three keys from three separate AISStream accounts.** AISStream limits connections **per account, not per key**. The app opens three independent streams — **area monitoring** (AIS_API_KEY), **followed ships** (FOLLOW_AIS_API_KEY), and **coverage map** (HEATMAP_AIS_API_KEY). If two use keys from the **same account**, they compete for the same slot and one keeps getting rejected. Give each feature a key from a **dedicated account**.

app.config.properties — operating parameters

Tracker

Monitoraggi

Navi seguite

Mappa zone coperte

Avvia il monitoraggio

Ferma

Aree

Impostazioni

Notifiche

GNV ORION spostata da — a toscana 12:52:23 28/07/26

Area: Toscana — attiva

Manuale amministratore

Manuale utente

Log attività

494 letture

Tracking porti - Toscana

ATTIVO

Indietro

GeneraliNotificheAreeIntegrazioni esterneParametriBackup / RipristinoLog attivitàLog APIDiagnostica AIS

Le modifiche a questi parametri vengono scritte in app.config.properties e richiedono il riavvio del server per avere effetto (non basta ricaricare il browser). I segreti e le credenziali restano nel file local.properties ; i toggle di import e notifiche sono nel tab Generali.

Salva parametri2 modifiche non salvate

RILEVAMENTO STATO NAVE

Soglia SOG (nodi) sotto cui la nave è considerata ferma

SOG_FERMA_KN

0,5

kn

Raggio (metri): ultime posizioni entro questo raggio → nave ormeggiata (isteresi anti-swing)

STILL_RADIUS_M

100

m

FINESTRE DI VISIBILITÀ

Ore: una nave in movimento viene mostrata in "Navi presenti" se vista in questa finestra

ACTIVE_WINDOW_HOURS

6

ore

Ore: una nave in porto (ormeggiata/all'ancora/ferma) resta in "Navi presenti" più a lungo

PORT_WINDOW_HOURS

24

ore

CONNESSIONE E POLLING

Millisecondi: attesa prima di riconnettersi ad AISStream dopo un errore

RECONNECT_DELAY_MS

5000

ms

Millisecondi: intervallo di aggiornamento del frontend (default 5 minuti)

POLL_INTERVAL_MS

300000

ms

Settings — Parameters: configuration fields grouped by category with description.

Contains thresholds and parameters (time windows, radii, retention, berths, score weights). Format KEY=value , each parameter documented by a comment in the file. **You can also edit them from the interface at ⚙ Settings → Parameters** (more convenient). Either way, you need to **restart the server** to apply them (read only at startup). Examples:

Key	Meaning	Default
<code>SOG_FERMA_KN</code>	Speed (knots) below which a ship is "stationary"	<code>0.5</code>
<code>ACTIVE_WINDOW_HOURS</code>	Hours a moving ship stays among the "current" ones	6
<code>PORT_WINDOW_HOURS</code>	Hours a ship in port stays among the "current" ones	24
<code>POLL_INTERVAL_MS</code>	Interface refresh interval (ms)	300000
<code>AIS_OUTAGE_CHECK</code>	Enables AIS outage detection	<code>true</code>
<code>AIS_OUTAGE_SILENCE_MIN</code>	Minutes without signals before querying the uptime monitor	10
<code>AIS_UPTIME_SELFHOST_URL</code>	URL of your own self-hosted uptime monitor instance (queried first)	<i>(empty)</i>
<code>AIS_UPTIME_URL</code>	URL of the public uptime	<code>https://aisuptime.buttermilkgreen.fyi</code>

Key	Meaning	Default
	monitor, used as a fallback	
MAX_READINGS_PER_TYPE	Maximum readings kept per message type	10000
BERTH_CLUSTER_EPS_M	Mooring-to-berth clustering radius (meters)	80
BERTH_MIN_PTS	Minimum nearby moorings to form a berth	3
BERTH_MIN_MOORINGS	Minimum moorings before characterizing a berth	10
BERTH_DOMINANT_PCT	Percentage a category must exceed to name the berth	60
BERTH_RECOMPUTE_MIN	Minutes between one automatic berth recomputation and the next	30
HEATMAP_GRID_DEG	Coverage map cell size, in	0.25

Key	Meaning	Default
	degrees (~28 km)	
HEATMAP_FLUSH_SEC	How often (seconds) counts are written to disk	10
GROUP_ACTIVITY_LOG_RETENTION_DAYS	Retention (days) of the group activity log (see above)	90
TRANSIT_STOP_MIN_H	Minimum hours in an area for a visit to count as a call rather than a mere crossing (transit-area search and area-change notification)	3
TRANSIT_STOP_MAX_SOG_KN	Minimum observed speed below which the ship counts as genuinely stopped (recent visits only, whose positions are still stored)	0.5
TRANSIT_MIN_KN	Minimum implied	4

Key	Meaning	Default
	average speed of a direct passage between two areas: below it, the elapsed time counts as too long	
TRANSIT_MIN_SLACK_H	Floor of the time limit, so nearby areas are not penalised	12
TRANSIT_MAX_GAP_DAYS	Cap of the time limit, whatever the distance	30
TRANSIT_MAX_ROWS	Maximum ships returned by one transit-area search	500
AREA_CHANGE_REQUIRE_STOP	Fire the area-change notification only if the ship called at the origin area	true

Key	Meaning	Default
AREA_CHANGE_REQUIRE_PLAUSIBLE_TIME	Fire it only if the origin call is recent enough to explain the arrival	true
AREA_CHANGE_SKIP_OVERLAPPING	No area-change notification between two areas whose boxes overlap	true
RISK_*	Risk score weights and thresholds (see comments in the file)	various

bounding-boxes.json — area definitions

Lists the monitoring areas. **The recommended way to manage them is the  Areas screen** (which rewrites this file on its own). You can edit it by hand for initial provisioning; if so, **restart** the app.

Each area:

```
"bari": { "name": "Porto di Bari", "keyword": "BARI", "sw": [40.95, 16.60], "ne": [41.30, 1
```

Field	Meaning
key (bari)	Internal area identifier (also used by BBOX_PRESET)
name	Name shown in the interface
keyword	(Optional, can be null) filters “Expected ships” by destination
sw	South-West corner [lat, lon] in decimal degrees
ne	North-East corner [lat, lon] in decimal degrees

Manual edits to this file while the app is running only take effect after a restart. If you use the Areas screen, the file’s formatting gets normalized (stays valid, but indentation changes).

Backup, restore, and deploy

The screenshot shows the 'Tracker' application interface. On the left is a sidebar with navigation options: Monitoraggi, Navi seguite, Mappa zone coperte, Avvia il monitoraggio, Ferma, Aree, Impostazioni, Notifiche, GNV ORION (spostata da toscana), Area: Toscana, Manuale amministratore, Manuale utente, and Log attività. The main panel is titled 'Tracking porti - Toscana' and has a status 'ATTIVO'. It contains several tabs: Generali, Notifiche, Aree, Integrazioni esterne, Parametri, Backup / Ripristino (selected), Log attività, Log API, and Diagnostica AIS. The 'Backup / Ripristino' tab is divided into three sections: 'AUTO-BACKUP LOCALE' with a list of automatic backups (each with a download icon and a 'Ripristina' button), 'BACKUP COMPLETO' with 'Esporta tutto' and 'Importa tutto' buttons, and 'DATI' with 'Esporta CSV' and 'Backup database' buttons. The top right of the interface shows user roles: Amministratore, Admin, and Esci.

Settings — Backup tab: download and export data, restore a backup.

From **Settings** → **Backup** (visible only to administrators) you **download a backup** of the database, **restore** a saved backup, and **export** data.

- **Restoring** a database replaces **all** current data (irreversible): download a backup first. After restoring, data is reassigned to the correct area based on coordinates. Restoring **does not** re-trigger VesselFinder/MarineTraffic scraping (the data is already in the restored DB).
- **Coverage map** data lives in a **separate database**, exportable/importable on its own and still included in the full backup.
- **Auto-restore after a deploy**: the database is wiped whenever the application is updated. If, at startup, the DB **doesn't exist** and there are **auto-backups** present (`data/backups/` folder), the app automatically restores the latest backup. Requires the backups folder to survive the deploy. Doesn't trigger if the DB exists but was simply emptied with "Clear data". Can be disabled with `AUTO_RESTORE_ON_DEPLOY=false` in `app.config.properties`.

Credits

AIS outage detection (the outage banner) relies on the **AISStream-Uptime** project by buttermilkgreen, an uptime monitor for the AISStream service. The app doesn't embed its code: it only queries the public API of its hosted instance (<https://aisuptime.buttermilkgreen.fyi>) to figure out whether a data silence is due to a service outage or simply a quiet area. The project is open source (MIT license) and you can **self-host it**: point `AIS_UPTIME_SELFHOST_URL` at your own instance's URL.